

TidyTuesday Week 40

Pranay Gundam

Saturday 3rd October, 2026

Contents

1	Weekly Summary	2
2	Date: 2026-09-28	4
2.1	Raw Regression	4
2.2	Detrended Regression	6
3	Date: 2026-09-29	8
3.1	Raw Regression	8
3.2	Detrended Regression	10
4	Date: 2026-09-30	12
4.1	Raw Regression	12
4.2	Detrended Regression	14
5	Date: 2026-10-01	16
5.1	Raw Regression	16
5.2	Detrended Regression	18
6	Date: 2026-10-02	20
6.1	Raw Regression	20
6.2	Detrended Regression	22
7	Date: 2026-10-03	24
7.1	Raw Regression	24
7.2	Detrended Regression	26

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we feed a robot two completely unrelated economic time series, demand it find meaning, and then act shocked when it does. This week's contestants included Great Lakes wood product manufacturing, European cell phone subscriptions, discontinued labor force shares, intellectual property import fees, computer training price indexes, and – a personal favorite – the Import Price Index for Beverages and Tobacco. Nothing says rigorous science like regressing booze-and-cigarette prices against unfilled orders for camera parts. Buckle up.

Let's start with the week's trend-mirage hall of fame, Tuesday the 29th, where Depression-era wage-and-salary income shares were pitted against the number of Black Americans not in the labor force. The raw regression came back with an R-squared of 0.89 and a p-value with so many zeros after the decimal point it looks like a typo. Economists would weep. But then we detrend it, strip out the fact that both lines simply wander across the twentieth century, and the R-squared face-plants from 0.89 all the way down to 0.23. Still technically significant, sure, but mostly this is a textbook example of "two things drifting in the same era is not a relationship, it's a calendar." The computer-training price index versus finance company securities flows (the 2nd) pulled the same stunt in reverse: dead-on-arrival in the raw version, then suddenly perking up to a flirty p-value of 0.07 once detrended, which is the statistical equivalent of a result clearing its throat and saying "well, maybe."

Now, the one that actually deserves a raised eyebrow: Monday the 28th, wood product manufacturing in the Great Lakes versus European cell phone subscriptions. The raw regression was a nothingburger – not significant, barely any explanatory power. But detrended, it got stronger, landing at an R-squared of 0.18 and a genuinely significant p-value of 0.02. That's the rare and precious case where a relationship shows up only after you remove the shared drift, meaning it survives the one test this whole project exists to administer. Do I have a theory for why Midwestern lumber output moves inversely with how many phones Europeans own? I do not. I will not be developing one. But it held up, so it gets the gold star.

The rest of the week was mercifully honest about its own pointlessness: Thursday's government imputations versus official reserve assets couldn't find a signal with a flashlight (R-squared of 0.0009, truly inspiring), and Saturday's tobacco-vs-cameras pairing looked significant raw then evaporated to a p-value of 0.76 once detrended, which is just the universe politely confirming nothing was there. As always, the standing reminder: these are randomly paired series, no causal claims are being made, and spurious correlation isn't the risk here – it's the entire business model.

See you next week for more statistically significant nonsense.

2 Date: 2026-09-28

Series ID: GLAKWPMANRGSP

This series is titled Real Gross Domestic Product: Wood Product Manufacturing (321) in the Great Lakes BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: ITCELSETSP2ECS

This series is titled Mobile Cellular Subscriptions: All Income Levels in Europe and Central Asia and has a frequency of Annual. The units are Number per 100 People and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2025-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITCELSETSP2ECS	R-squared:	0.083
Model:	OLS	Adj. R-squared:	0.048
Method:	Least Squares	F-statistic:	2.357
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.137
Time:	18:50:21	Log-Likelihood:	-142.11
No. Observations:	28	AIC:	288.2
Df Residuals:	26	BIC:	290.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	285.3207	123.888	2.303	0.030	30.665	539.976
value_fred_GLAKWPMANRGSP	-0.0407	0.026	-1.535	0.137	-0.095	0.014

Omnibus:	3.492	Durbin-Watson:	0.174
Prob(Omnibus):	0.174	Jarque-Bera (JB):	2.983
Skew:	-0.713	Prob(JB):	0.225
Kurtosis:	2.275	Cond. No.	7.63e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 7.63e+04. This might indicate that there are strong multicollinearity or other numerical problems.

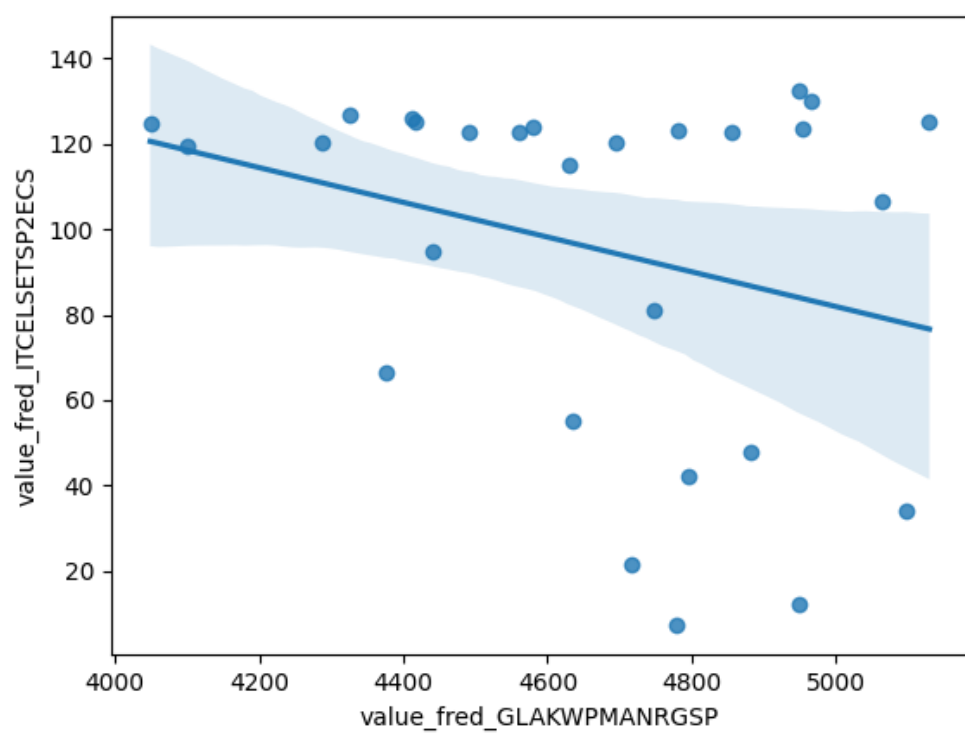


Figure 1: Raw Regression Plot for 2026-09-28

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITCELSETSP2ECS_detrended	R-squared:	0.184
Model:	OLS	Adj. R-squared:	0.152
Method:	Least Squares	F-statistic:	5.853
Date:	Mon, 28 Sep 2026	Prob (F-statistic):	0.0228
Time:	18:50:21	Log-Likelihood:	-119.03
No. Observations:	28	AIC:	242.1
Df Residuals:	26	BIC:	244.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.116e-12	3.330	3.35e-13	1.000	-6.845	6.845
value_fred_GLAKWPMANRGSP_detrended	-0.0283	0.012	-2.419	0.023	-0.052	-0.004

Omnibus:	0.861	Durbin-Watson:	0.444
Prob(Omnibus):	0.650	Jarque-Bera (JB):	0.837
Skew:	0.216	Prob(JB):	0.658
Kurtosis:	2.272	Cond. No.	285.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

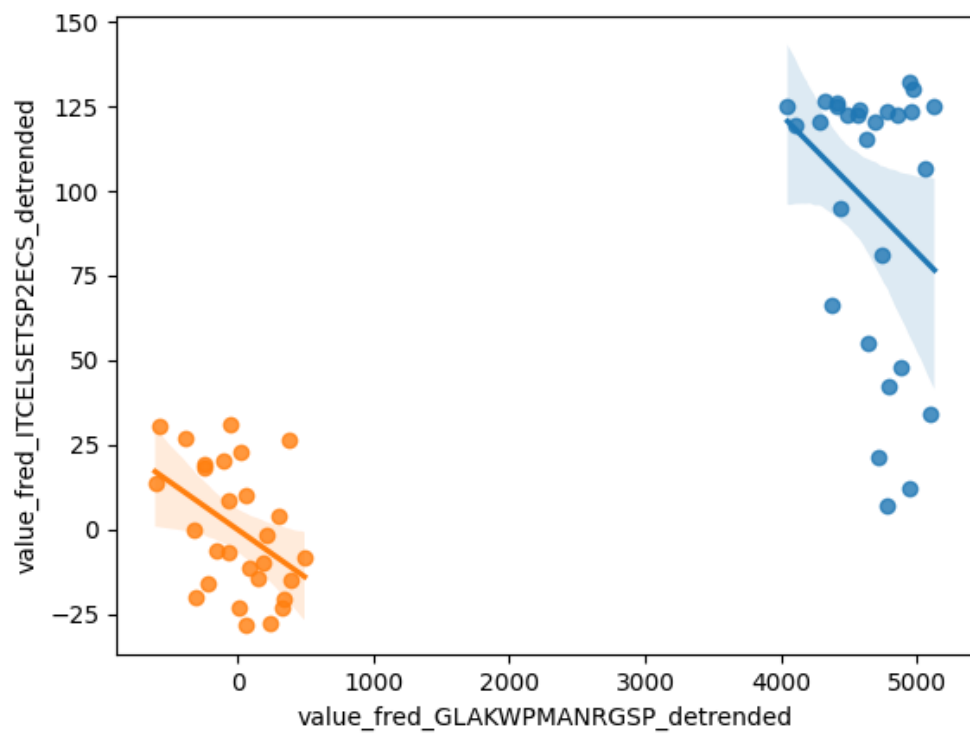


Figure 2: Detrended Regression Plot for 2026-09-28

3 Date: 2026-09-29

Series ID: W269RE1A156NBEA

This series is titled Shares of gross domestic income: Compensation of employees, paid: Wage and salary accruals: Disbursements (DISCONTINUED) and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2011-01-01. The popularity of this series is 1.

Series ID: LNU05000006

This series is titled Not in Labor Force, Black or African American and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1972-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LNU05000006	R-squared:	0.886			
Model:	OLS	Adj. R-squared:	0.883			
Method:	Least Squares	F-statistic:	295.5			
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	1.63e-19			
Time:	17:07:32	Log-Likelihood:	-300.21			
No. Observations:	40	AIC:	604.4			
Df Residuals:	38	BIC:	607.8			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t 	[0.025	0.975]
const	3.833e+04	1754.440	21.850	0.000	3.48e+04	4.19e+04
value_fred_W269RE1A156NBEA	-632.0219	36.768	-17.190	0.000	-706.454	-557.590
Omnibus:	2.321	Durbin-Watson:	0.414			
Prob(Omnibus):	0.313	Jarque-Bera (JB):	1.433			
Skew:	0.178	Prob(JB):	0.488			
Kurtosis:	2.143	Cond. No.	1.17e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.17e+03. This might indicate that there are strong multicollinearity or other numerical problems.

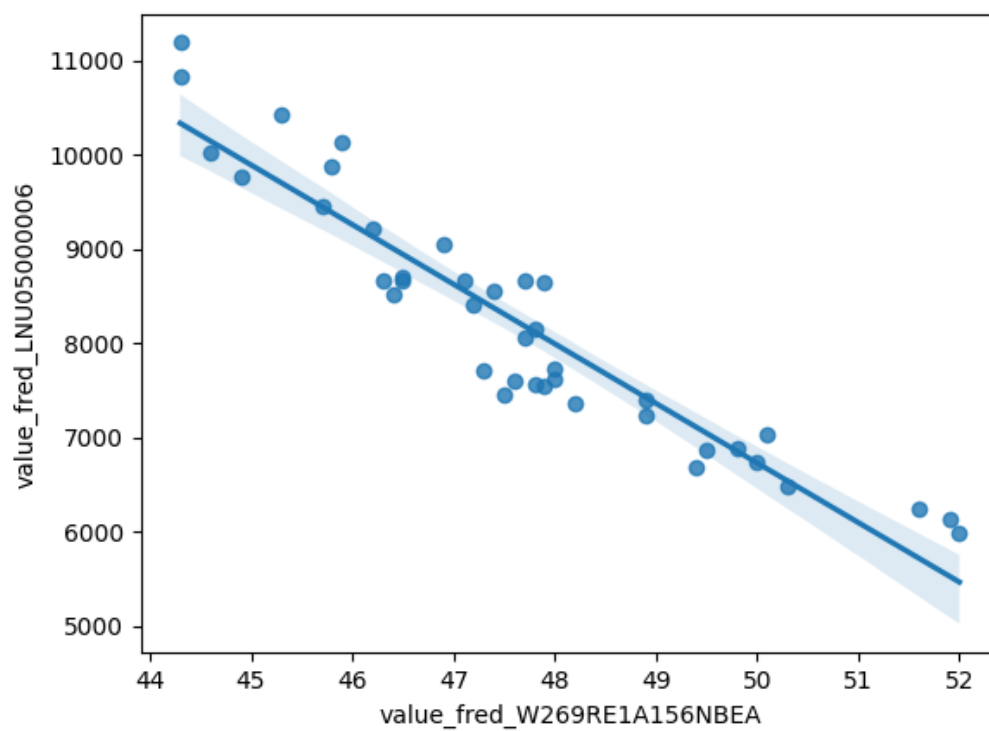


Figure 3: Raw Regression Plot for 2026-09-29

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LNU05000006_detrended	R-squared:	0.229
Model:	OLS	Adj. R-squared:	0.209
Method:	Least Squares	F-statistic:	11.29
Date:	Tue, 29 Sep 2026	Prob (F-statistic):	0.00178
Time:	17:07:32	Log-Likelihood:	-275.80
No. Observations:	40	AIC:	555.6
Df Residuals:	38	BIC:	559.0
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.404e-11	38.755	1.14e-12	1.000	-78.455	78.455
value_fred_W269RE1A156NBEA_detrended	-174.7065	51.991	-3.360	0.002	-279.957	-69.456

Omnibus:	8.056	Durbin-Watson:	0.312
Prob(Omnibus):	0.018	Jarque-Bera (JB):	6.850
Skew:	0.862	Prob(JB):	0.0325
Kurtosis:	4.067	Cond. No.	1.34

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

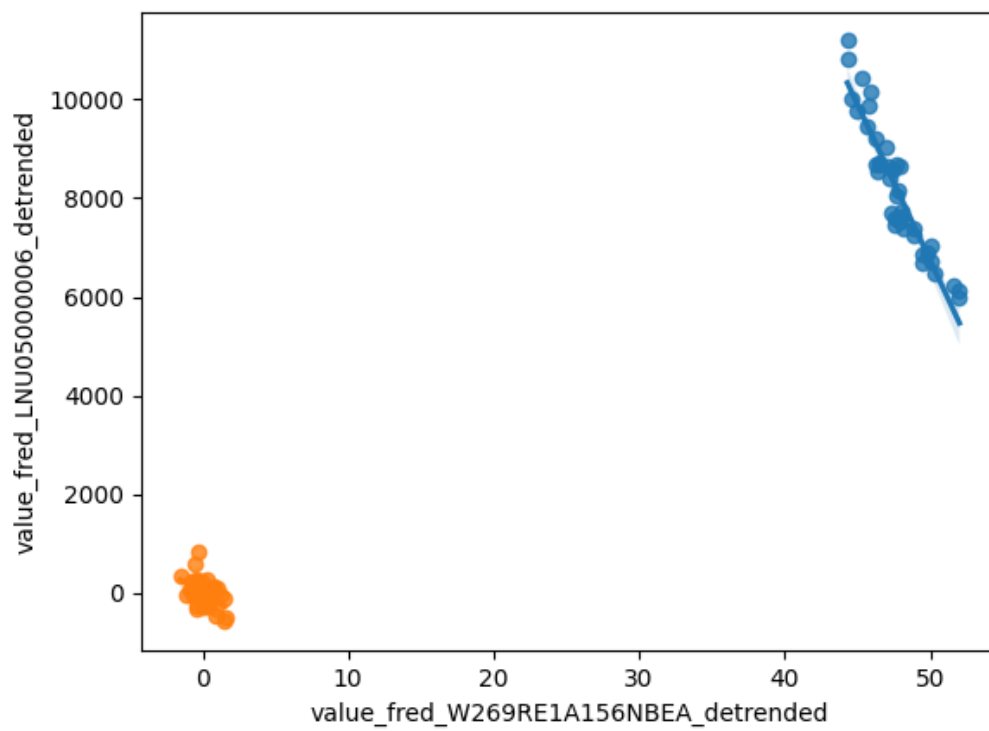


Figure 4: Detrended Regression Plot for 2026-09-29

4 Date: 2026-09-30

Series ID: IEAMSIPN

This series is titled Imports of Services: Charges for the use of intellectual property n.i.e. and has a frequency of Quarterly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: DDURRE1A156NBEA

This series is titled Shares of gross domestic product: Personal consumption expenditures: Durable goods and has a frequency of Annual. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2025-01-01. The popularity of this series is 4.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DDURRE1A156NBEA	R-squared:	0.321
Model:	OLS	Adj. R-squared:	0.294
Method:	Least Squares	F-statistic:	11.82
Date:	Wed, 30 Sep 2026	Prob (F-statistic):	0.00206
Time:	17:05:45	Log-Likelihood:	-25.239
No. Observations:	27	AIC:	54.48
Df Residuals:	25	BIC:	57.07
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	8.7930	0.314	28.002	0.000	8.146	9.440
value_fred_IEAMSIPN	-0.0001	3.39e-05	-3.438	0.002	-0.000	-4.67e-05

Omnibus:	5.919	Durbin-Watson:	0.272
Prob(Omnibus):	0.052	Jarque-Bera (JB):	1.819
Skew:	0.005	Prob(JB):	0.403
Kurtosis:	1.729	Cond. No.	2.36e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.36e+04. This might indicate that there are strong multicollinearity or other numerical problems.

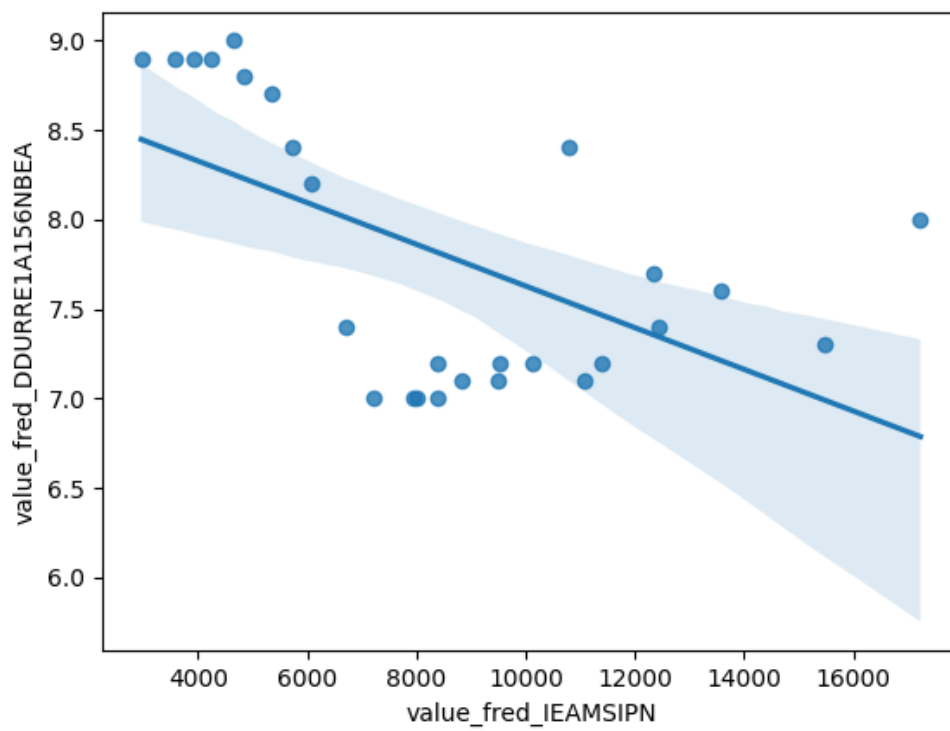


Figure 5: Raw Regression Plot for 2026-09-30

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DDURRE1A156NBEA_detrended	R-squared:	0.024
Model:	OLS	Adj. R-squared:	-0.015
Method:	Least Squares	F-statistic:	0.6106
Date:	Wed, 30 Sep 2026	Prob (F-statistic):	0.442
Time:	17:05:45	Log-Likelihood:	-23.256
No. Observations:	27	AIC:	50.51
Df Residuals:	25	BIC:	53.10
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	6.162e-14	0.115	5.38e-13	1.000	-0.236	0.236
value_fred_IEAMSIPN_detrended	8.159e-05	0.000	0.781	0.442	-0.000	0.000

Omnibus:	0.623	Durbin-Watson:	0.358
Prob(Omnibus):	0.732	Jarque-Bera (JB):	0.652
Skew:	-0.067	Prob(JB):	0.722
Kurtosis:	2.250	Cond. No.	1.10e+03

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.1e+03. This might indicate that there are strong multicollinearity or other numerical problems.

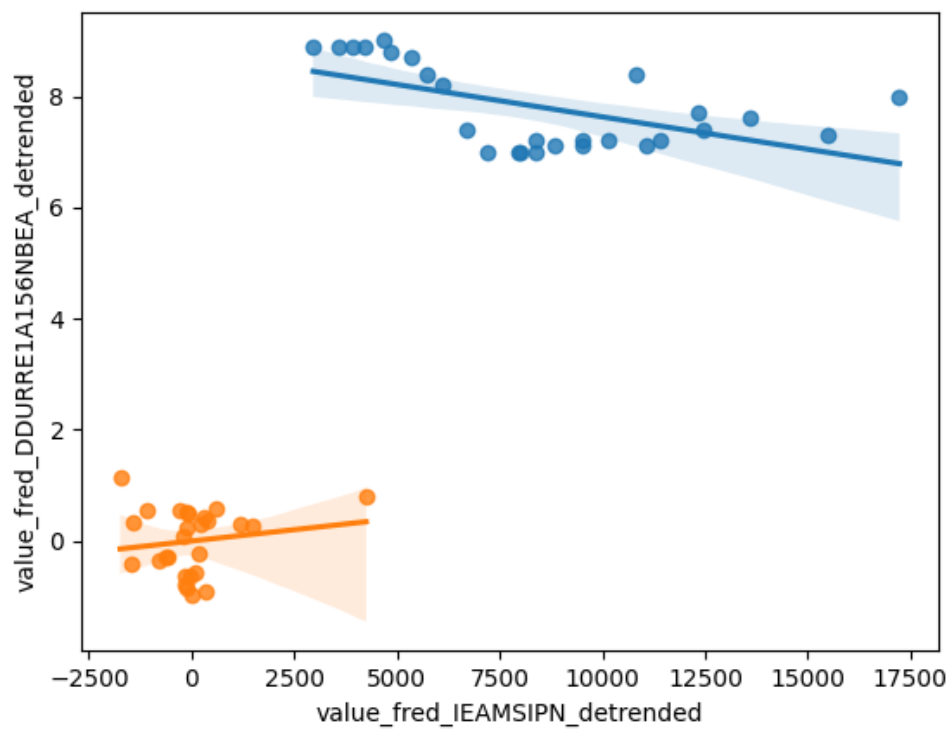


Figure 6: Detrended Regression Plot for 2026-09-30

5 Date: 2026-10-01

Series ID: W568RC1A027NBEA

This series is titled Government current receipts: Imputations and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2022-01-01. The popularity of this series is 1.

Series ID: BOPOON

This series is titled U.S. Official Reserve Assets Abroad, Net (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1960-01-01 and the observation end date is 2014-01-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BOPOON	R-squared:	0.001
Model:	OLS	Adj. R-squared:	-0.018
Method:	Least Squares	F-statistic:	0.04727
Date:	Thu, 01 Oct 2026	Prob (F-statistic):	0.829
Time:	19:55:13	Log-Likelihood:	-112.57
No. Observations:	55	AIC:	229.1
Df Residuals:	53	BIC:	233.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-0.2012	0.458	-0.439	0.662	-1.119	0.717
value_fred_W568RC1A027NBEA	-0.0115	0.053	-0.217	0.829	-0.117	0.094

Omnibus:	6.566	Durbin-Watson:	1.473
Prob(Omnibus):	0.038	Jarque-Bera (JB):	10.848
Skew:	0.067	Prob(JB):	0.00441
Kurtosis:	5.172	Cond. No.	15.6

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

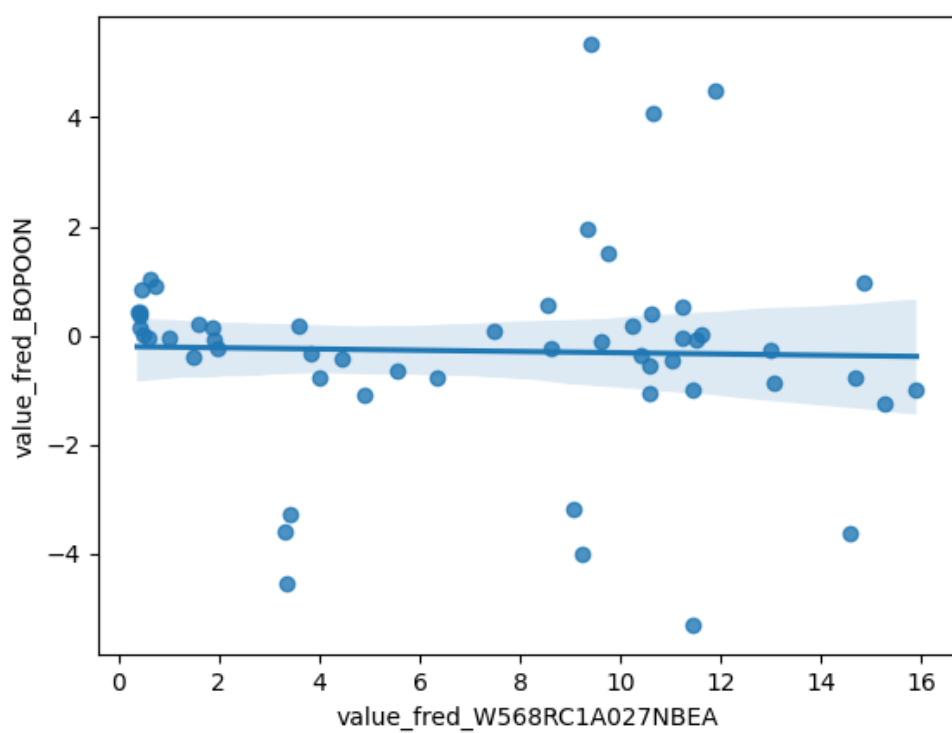


Figure 7: Raw Regression Plot for 2026-10-01

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BOPOON_detrended	R-squared:	0.007
Model:	OLS	Adj. R-squared:	-0.011
Method:	Least Squares	F-statistic:	0.3913
Date:	Thu, 01 Oct 2026	Prob (F-statistic):	0.534
Time:	19:55:13	Log-Likelihood:	-112.39
No. Observations:	55	AIC:	228.8
Df Residuals:	53	BIC:	232.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.084e-14	0.256	-4.22e-14	1.000	-0.514	0.514
value_fred_W568RC1A027NBEA_detrended	-0.1030	0.165	-0.626	0.534	-0.433	0.227

Omnibus:	5.994	Durbin-Watson:	1.499
Prob(Omnibus):	0.050	Jarque-Bera (JB):	9.174
Skew:	0.019	Prob(JB):	0.0102
Kurtosis:	5.000	Cond. No.	1.56

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



6 Date: 2026-10-02

Series ID: PCU6114206114201

This series is titled Producer Price Index by Industry: Computer Training: Computer Training Services and has a frequency of Monthly. The units are Index Jun 2006=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-06-01 and the observation end date is 2017-08-01. The popularity of this series is 1.

Series ID: STFAFSXDFBANQ

This series is titled Domestic Finance Companies, Securities, Flow and has a frequency of Quarterly. The units are Millions of Dollars, Quarterly Rate and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2011-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_STFAFSXDFBANQ	R-squared:	0.019
Model:	OLS	Adj. R-squared:	-0.020
Method:	Least Squares	F-statistic:	0.4905
Date:	Fri, 02 Oct 2026	Prob (F-statistic):	0.490
Time:	16:53:47	Log-Likelihood:	-262.34
No. Observations:	27	AIC:	528.7
Df Residuals:	25	BIC:	531.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	2.662e+04	3.85e+04	0.691	0.496	-5.28e+04	1.06e+05
value_fred_PCU6114206114201	-233.6407	333.610	-0.700	0.490	-920.724	453.443

Omnibus:	10.482	Durbin-Watson:	1.690
Prob(Omnibus):	0.005	Jarque-Bera (JB):	9.849
Skew:	1.008	Prob(JB):	0.00726
Kurtosis:	5.166	Cond. No.	5.55e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 5.55e+03. This might indicate that there are strong multicollinearity or other numerical problems.

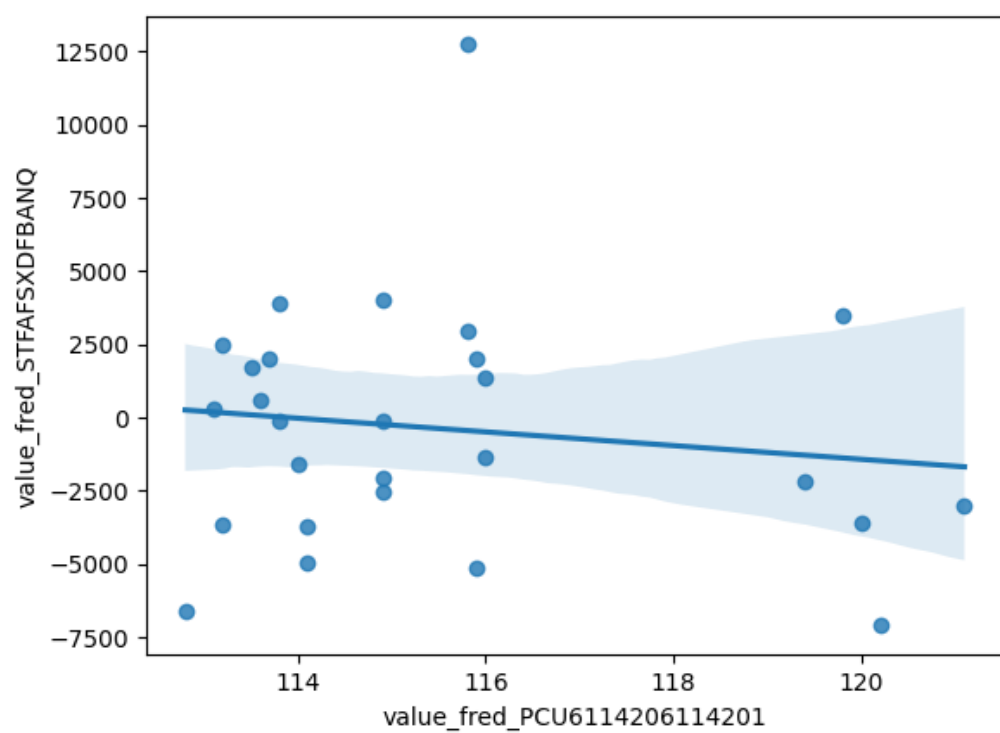


Figure 9: Raw Regression Plot for 2026-10-02

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_STFAFSXDFBANQ_detrended	R-squared:	0.123
Model:	OLS	Adj. R-squared:	0.088
Method:	Least Squares	F-statistic:	3.501
Date:	Fri, 02 Oct 2026	Prob (F-statistic):	0.0731
Time:	16:53:47	Log-Likelihood:	-260.81
No. Observations:	27	AIC:	525.6
Df Residuals:	25	BIC:	528.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.618e-08	758.492	-2.13e-11	1.000	-1562.143	1562.143
value_fred_PCU6114206114201_detrended	-1205.5548	644.324	-1.871	0.073	-2532.565	121.455

Omnibus:	3.116	Durbin-Watson:	1.754
Prob(Omnibus):	0.211	Jarque-Bera (JB):	1.662
Skew:	0.522	Prob(JB):	0.436
Kurtosis:	3.624	Cond. No.	1.18

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

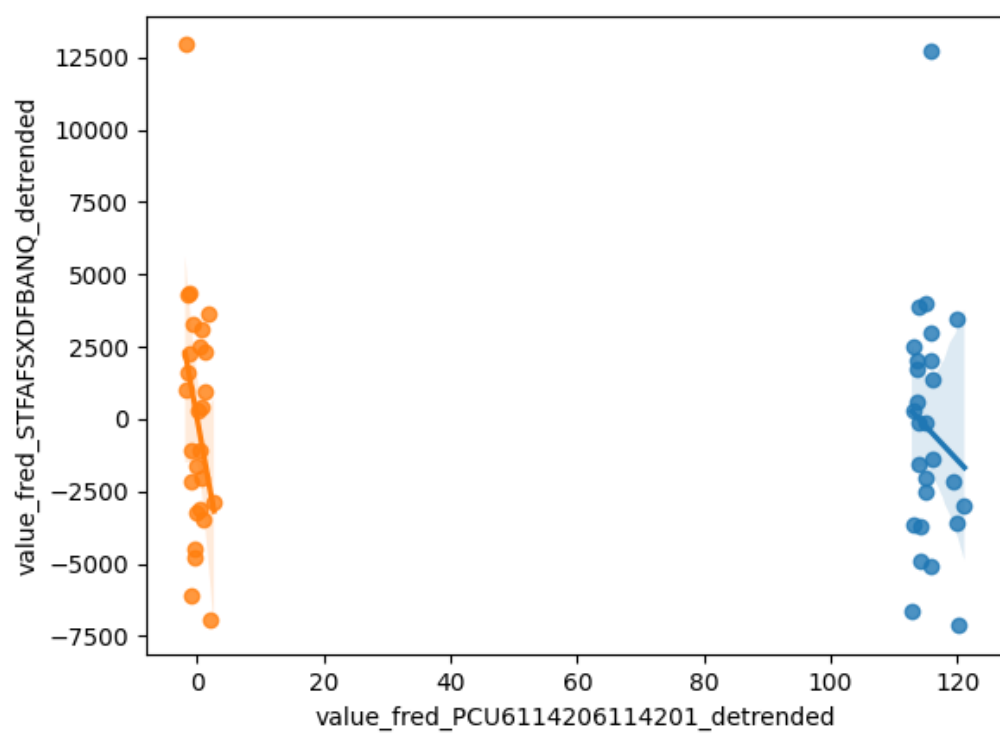


Figure 10: Detrended Regression Plot for 2026-10-02

7 Date: 2026-10-03

Series ID: IM1

This series is titled Import Price Index (SITC): Beverages and tobacco (DISCONTINUED) and has a frequency of Monthly. The units are Index 2000=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1995-01-01 and the observation end date is 2005-12-01. The popularity of this series is 1.

Series ID: A33GUO

This series is titled Manufacturers' Unfilled Orders: Photographic Equipment Manufacturing and has a frequency of Monthly, End of Period. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1992-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

7.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_A33GUO	R-squared:	0.048			
Model:	OLS	Adj. R-squared:	0.041			
Method:	Least Squares	F-statistic:	6.572			
Date:	Sat, 03 Oct 2026	Prob (F-statistic):	0.0115			
Time:	15:18:33	Log-Likelihood:	-929.10			
No. Observations:	132	AIC:	1862.			
Df Residuals:	130	BIC:	1868.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-61.5812	429.804	-0.143	0.886	-911.897	788.734
value_fred_IM1	11.0522	4.311	2.564	0.011	2.523	19.582
Omnibus:	1.052	Durbin-Watson:	0.033			
Prob(Omnibus):	0.591	Jarque-Bera (JB):	0.646			
Skew:	0.122	Prob(JB):	0.724			
Kurtosis:	3.241	Cond. No.	1.77e+03			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.77e+03. This might indicate that there are strong multicollinearity or other numerical problems.

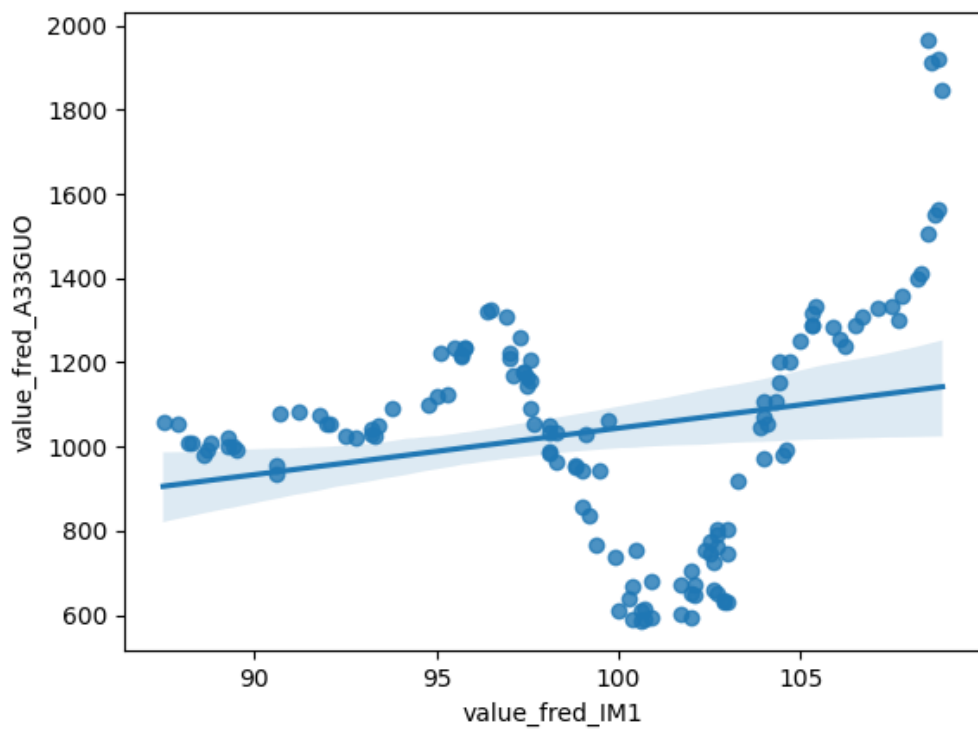


Figure 11: Raw Regression Plot for 2026-10-03

7.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_A33GUO_detrended	R-squared:	0.001
Model:	OLS	Adj. R-squared:	-0.007
Method:	Least Squares	F-statistic:	0.09245
Date:	Sat, 03 Oct 2026	Prob (F-statistic):	0.762
Time:	15:18:33	Log-Likelihood:	-929.09
No. Observations:	132	AIC:	1862.
Df Residuals:	130	BIC:	1868.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.762e-10	24.188	-7.28e-12	1.000	-47.854	47.854
value_fred_IM1_detrended	7.4297	24.436	0.304	0.762	-40.913	55.773

Omnibus:	1.009	Durbin-Watson:	0.033
Prob(Omnibus):	0.604	Jarque-Bera (JB):	0.617
Skew:	0.125	Prob(JB):	0.735
Kurtosis:	3.223	Cond. No.	1.01

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

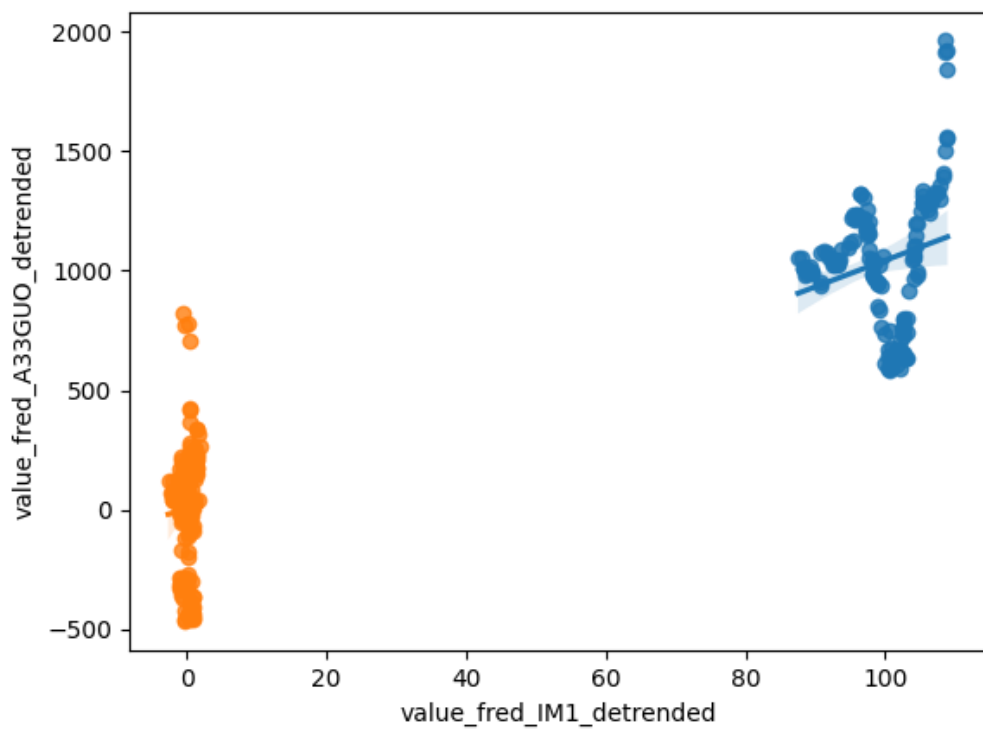


Figure 12: Detrended Regression Plot for 2026-10-03